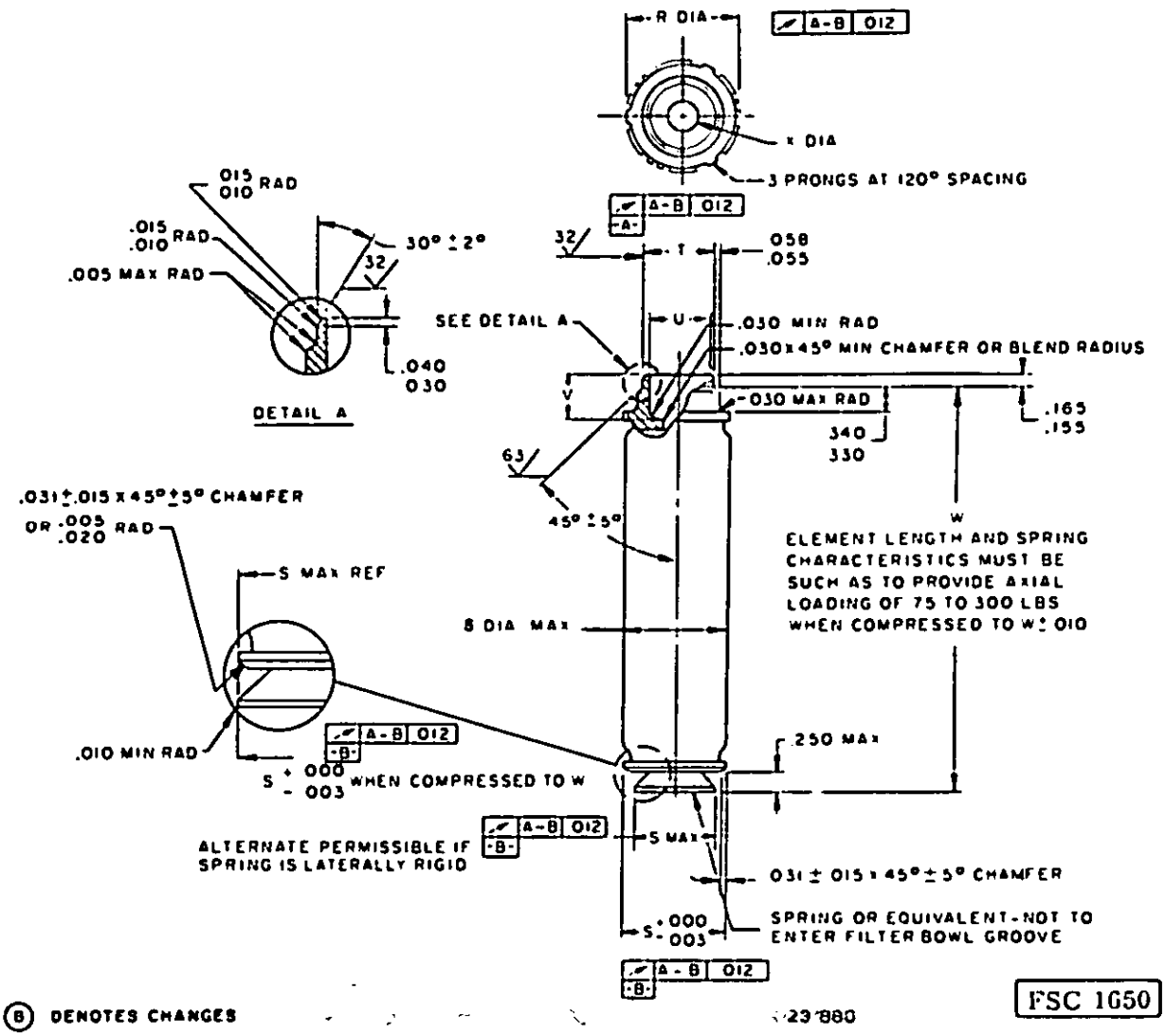
MILITARY SPECIFICATION SHEET

FILTER ELEMENT, FLUID, PRESSURE, HYDRAULIC LINE, 3000 PSI, ABSOLUTE 15 MICRONS, -65° F TO +275° F

This specification is approved for use by all Department and Agencies of the Department of Defense.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-8815.



FSC 1650

(B) DENOTES CHANGES

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NOTES: 1. Dimensions in inches unless otherwise specified. See above Figure and Table A.

2. Tolerances, decimals ±.010, angles ±1°.

3. Finish all over 100 except as noted. Surface texture in accordance with ASA B46.1 - 1962 (ANSI).

4. Material shall be in accordance with the procurement specification.

5. Filter element for use in M8815/1 and M8815/2 filter assemblies.

TABLE A

DASH NO.		RATED MAX FLOW GPM	R +.000 -.010	S	T ±.001	U MIN	V MIN	W	X MIN	MAX WT LBS
-8	-8C	6.0	1.250	1.137	.748	.670	.490	4.095	.435	0.35
-10	-10C	10.5	1.415	1.290	.875	.790	.600	5.235	.555	0.55
-12	-12C	16.0	1.625	1.500	1.122	1.035	.690	6.235	.715	0.83
-16	-16C	29.0	1.985	1.845	1.372	1.155	.760	7.235	.995	1.50

REQUIREMENTS:

Pressure Drop: Maximum pressure drop at rated flow across the filter element shall not exceed 10 psid.

Degree of filtration: The filter element shall remove 100 percent of all contaminant larger than 15 microns and 94 percent of all test contaminants. The filter element efficiency shall be a minimum of 94 percent.

Element Collapse: The filter element shall withstand a differential pressure of 4500 psid.

(B) Cold Start: Noncleanable element shall withstand 3000 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop shall not exceed 10 psid following the cold start test.

Maximum Particle Passed: The maximum particle passed shall be no greater than 15 microns.

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Flow Fatigue: The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cpm maximum.

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
500 cycles at 3000 psid pressure drop

Part Number: The part number consists of the basic number of this specification sheet and a dash number taken from Table A. Add letter C after dash number to indicate cleanable filter element.

Example:

M8815/3-8	Noncleanable filter element.
M8815/3-8C	Cleanable filter element.
M8815/3-16C	Cleanable filter element.

TABLE I. CONTAMINANTS

Dash No.	Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup			Migration Analysis Value
		F-9 Beads (gms)	AC Dust (gms)	F-9 Beads (gms)	Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms) 1/	Minimum AC Dust to 90 psid (gms) 2/	Maximum Allowable (gms)
-8	6.0	0.40	-	0.05	-	0.12	1.80	0.6	0.0015
-10	10.5	0.70	-	0.05	-	0.21	3.15	1.05	0.0026
-12	16.0	1.05	-	0.05	-	0.32	4.80	1.60	0.0040
-16	29.0	1.45	-	0.05	-	0.44	6.60	2.20	0.0073

(B)

1/ For noncleanable filter elements
(B) 2/ For cleanable filter elements

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Intended Use: The filter elements covered by this specification sheet are intended for use as specified in paragraph 6.1 of the general specification.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Reviewer activities:
Army - MI
Air Force - 71

Preparing activity:
Navy - AS

(Project No. 1650-3362)

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